Ethan Mahajan

ENGINEERING PORTFOLIO

Mechanical engineering student focused on CAD, manufacturing, prototyping, and hands-on problem solving.

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From working prototypes to current design work

These five projects connect CAD, fabrication, electronics, and engineering communication. Each summary identifies my contribution, the evidence available, and the work still pending. The linked case studies provide complete documentation and demonstrations.

Contents

Artorius: FDM 3D Printer 1

Adjustable Phone Filming Mount 1

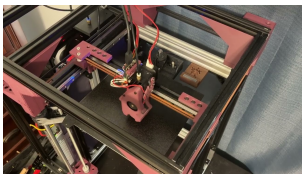
Air Purifier 2

Laser-Engraved Chess Board 2

Shape Machine 2

Artorius: FDM 3D Printer

Operational; refinement ongoing | Independent design and build



Physical printer, shown in the project video.

Design and contribution. Recurring problems with an older BIQU B1 led me to redesign its architecture around reusable motors, aluminum extrusion, and a 24 V supply. I developed a CoreXY frame, motion system, and custom toolhead in CAD, fabricated fixtures, assembled the machine, and integrated CAN-bus electronics with a serviceable enclosure.

Evidence and next steps. The printer is operational and the linked video shows Benchy printing. Motion tests confirmed 1,000 mm/s at 250,000 mm/s². An expected flow limit around 50 mm³/s gives an estimated 400 mm/s printing limit at 0.25 mm layer height and 0.5 mm line width. Cooling, print-quality tuning, and a full enclosure are next.

Tools: CAD, FDM fabrication, CoreXY motion, CAN bus

Full case study and supporting evidence

Adjustable Phone Filming Mount

Completed; usable first prototype | Independent design and fabrication



The usable first prototype: adjustable height, arm rotation, and phone orientation.

Design and contribution. I needed an inexpensive, adjustable way to film engineering builds with an iPhone. I modeled Fusion 360 fixtures for height, arm rotation, and phone orientation, printed them in ABS, and assembled the mount with aluminum extrusion, heat-set inserts, and fasteners. Chamfers and clearances supported FDM fabrication and assembly.

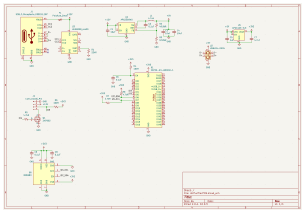
Evidence and next steps. The usable first prototype cost approximately $18 including extrusion, or $4 excluding it, as estimated in the project video. It became my project filming mount. Assembly exposed a missing screw-access hole; holder play and base shake suggest future stiffness improvements. A second build was unnecessary for the intended use.

Tools: Fusion 360, OrcaSlicer, ABS printing

Full case study and supporting evidence

Air Purifier

Design in progress | Independent mechanical and electronics design



KiCad schematic complete; PCB layout and mechanical design still undecided.

Design and contribution. I want to filter the air in my room and monitor particulate levels, including while 3D printing. The design uses a repurposed 120 mm fan, filter, particulate sensor, and custom controller. I expect to use laser-cut panels for the housing and printed fixtures.

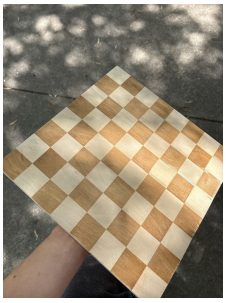
Evidence and next steps. The KiCad schematic is complete. PCB layout and mechanical design are still undecided, and the purifier has not been built.

Tools: KiCad, CAD; planned FDM printing and laser cutting

Full case study and supporting evidence

Laser-Engraved Chess Board

Physical board documented; metal pieces pending | Independent design and fabrication



Physical laser-engraved board; the CNC-machined metal pieces have not been made.

Design and contribution. I am developing a coordinated chess set through Georgia Tech's Invention Studio, combining a laser-engraved board with planned CNC-machined metal pieces. The CAD design prioritizes a manufacturable playing surface and stable, low-center-of-mass pieces. Separating the board and piece stages lets fabrication progress incrementally.

Evidence and next steps. Photographs document the physical board's engraved playing surface and personalized reverse; a video records the laser engraving process. The metal pieces have not been made and the full set is pending. Developing the CNC manufacturing approach is the next stage.

Tools: CAD, laser engraving; planned CNC machining

Full case study and supporting evidence

Shape Machine

Contribution ongoing | Website content and documentation

Shape Computation Lab identity. The linked page distinguishes my role from the team's technology.

Design and contribution. Within an interdisciplinary team, I improve the existing Shape Computation Lab website through content review, error correction, template maintenance, and GitHub documentation in the team's enterprise repository. The goal is to make explanations of Shape Machine's specialized shape-computation technology clearer and more consistent for visitors with varied technical backgrounds.

Evidence and next steps. My contribution is ongoing and focuses on the website's existing content and presentation. The case study describes those responsibilities and links to demonstrations of the team's technology for context. Specific before-and-after contribution examples remain to be documented; I do not claim authorship of the underlying technology.

Tools: Content editing, templates, GitHub

Full case study and supporting evidence